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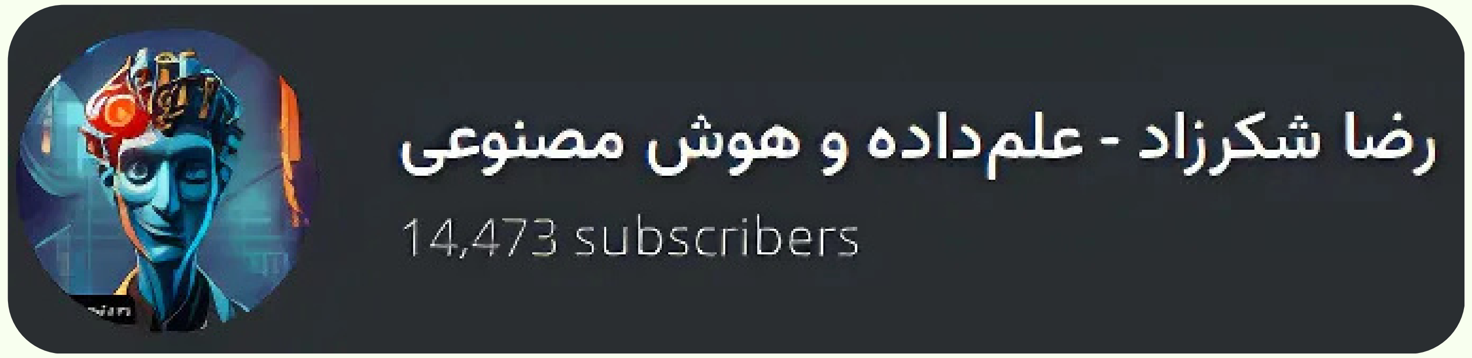
May 2025

NLP Workshop

Session 8 - Farsi NLP

Communities



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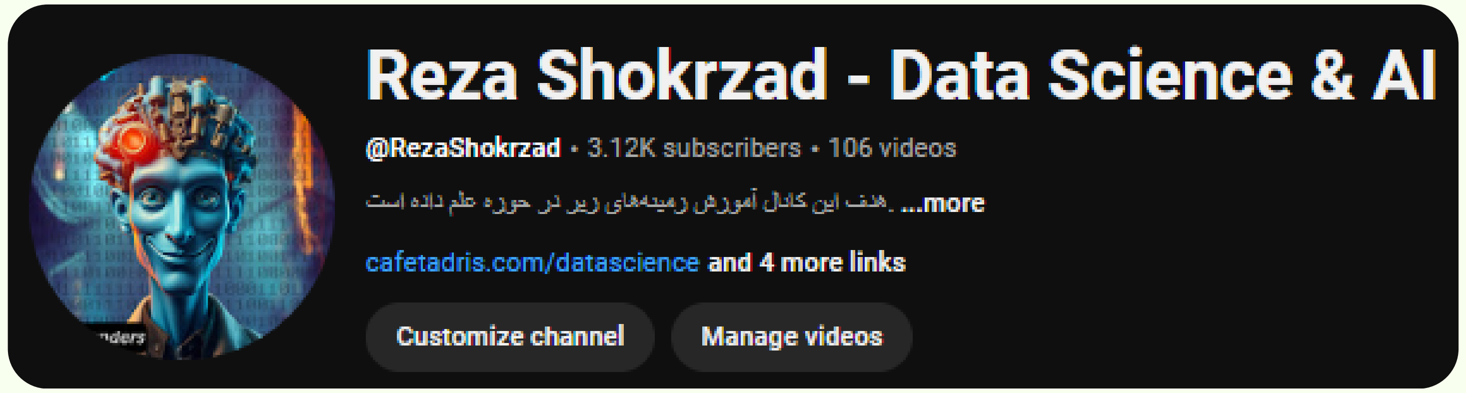

@RezaShokrzad

Content

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- Farsi Tokenization and Segmentation
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- Dependency Parsing Nuances
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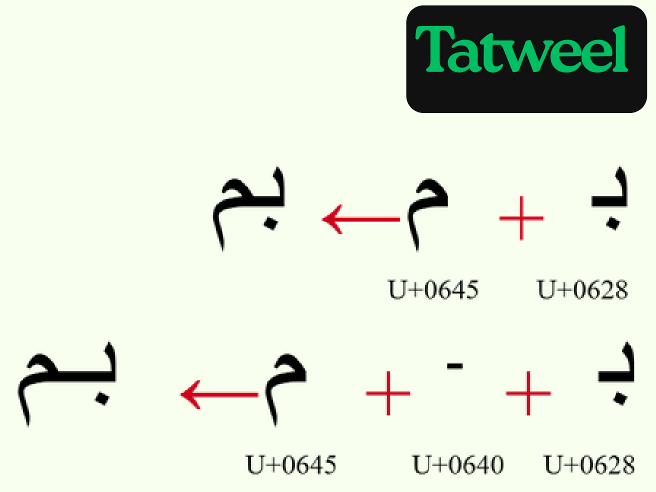
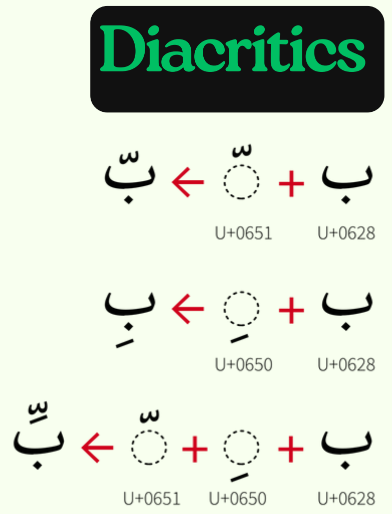
1. Script Normalization Basics

- **Same letter ≠ same code-point**
  - visually identical Arabic-vs-Persian glyphs carry different Unicode values **(ک vs ك)**
- **ZWNJ (zero-width non-joiner) & "half-space" breaks tokenizers**
  - inconsistent ZWNJ usage splits or fuses tokens unpredictably
- **Mixed Persian digits**
  - Persian «۱۲۳» and Western "123" must be mapped to one numeral set
- **Diacritics & Tatweel inflate vocab size**
  - optional vowel marks and elongation strokes add redundant tokens; strip to shrink vocab





https://arxiv.org/pdf/2111.03470

https://en.wikipedia.org/wiki/Zero-width_non-joiner

1. Script Normalization Core actions

- Canonical letters → map ك/ي → ک/ی
- Strip diacritics & Tatweel
- Digit unification → persian_digits=True
- Never drop ZWNJ inside words ("می‌رود")
- Collapse whitespace + preserve ZWNJ
- Standard punctuation ("," → "،")
- Roman-script ("Pinglish") needs separate step

```
>>> normalizer = Normalizer()
>>> normalizer.normalize('اصلاح نویسه ها و استفاده از نیم‌فاصله پردازش را آسان مي كند')
'اصلاح نویسه‌ها و استفاده از نیم‌فاصله پردازش را آسان می‌کند'
```

```
1 normalizer = Normalizer()
2 normalizer.normalize('مي رود 123٤٥۶')
```

```
'رود ۱۲۳۴۵۶\u200cمی'
```

Persian Rūmi	Perso-Arabic script	English
Yusefê gomgaşte báz áyad be Kanân qam maħor kolbeye ahzán şavad ruzi golestán qam maħor	یوسف گم گشته باز آید به کنعان غم مخور کلبه ی احزان شود روزی گلستان غم مخور	The lost Joseph will get back to Canaan, don't be sad The hut of madness will become a garden one day, don't be sad

http://en.wikipedia.org/wiki/Romanization_of_Persian#Fingilish

2. Tokenization & Segmentation

- **Affixed clitics: <کتابم = کتاب + ـم>**
  - Split bound pronouns/particles after segmentation to reveal true stem.
- **ZWNJ splits compounds: «می‌روم»**
  - Treat the half-space as a token break within multi-word verbs.
- **No caps → sentence starts ambiguous**
  - Detect sentence starts with punctuation cues instead of upper-case letters.
- **Numbers, Latin words, emojis in social text**
  - Design regex or subword rules to isolate digits, Latin text, and emojis.

https://arxiv.org/pdf/2202.10879

https://en.wikipedia.org/wiki/Clitic

2. Segmentation strategy

- Pre-normalise (slide 5 steps!)
- Detect half-spaces → keep as token boundary
- Rule hints for enclitics («ها», «تر», «مان» …)
- Fallback to sub-word (BPE/WordPiece) for OOV
- **HazmTokenizer** (regex + rules)
- **Stanza-fa** UDPipe models
- **spaCy-fa** (2024) with custom suffix matcher
- **BPE prep**: tokenizers.ByteLevelBPETokenizer on normalized corpus

```
>>> chunker = Chunker(model='chunker.model')
>>> tagged = tagger.tag(word_tokenize('کتاب خواندن را دوست داریم'))
>>> tree2brackets(chunker.parse(tagged))
'[کتاب خواندن NP] [را POSTP] [دوست داریم VP]'
```

```
from hazm import word_tokenize

sent = "کتابم را به او می‌دهم."
word_tokenize(sent)
```

```
['می', 'او', 'به', 'را', 'کتابم\u200cدهم', '.']
```

https://en.wikipedia.org/wiki/Byte_pair_encoding

3. Morphology & Lemmatization

- Rich person-tense inflection balloons vocab
- Suffix pronouns (+شان/+ت/+م) hide entities
- Accurate lemma boosts search & embedding recall
- Verbal chains: می+ROOT+ند/ه …
- Plural markers: ـها, ـان, ـین
- Compar./Superl.: -تر, -ترین
- Negation prefixes: نا-, بی-
- Derivational nouns: -ی, -گی

3. Morphology & Lemmatization

Processing steps

- Segmentation-aware stemming (slide 4 output)
- Morph-tag lookup (UD tags)
- Rule-backoff lemmatizer if OOV
- Post-check clitic detach

Tools / models

- Hazm stem/lemmatize, POS+morph tags
- Parsivar stemmer FindStems
- Stanza-fa: joint POS+lemma (UD 2.13)
- PerStem (light stemmer)

```python
from stanza import Pipeline
nlp = Pipeline('fa', processors='tokenize,pos,lemma')
doc = nlp("کتاب‌هایمان را می‌آورند")
[ (w.text, w.lemma) for w in doc.iter_words() ]
```

```
[('کتاب\u200cهای', 'کتاب'),
 ('مان', 'ما'),
 ('را', 'را'),
 ('می\u200cآورند', 'آورد')]
```

4. Dependency Parsing: Pipeline & Evaluation

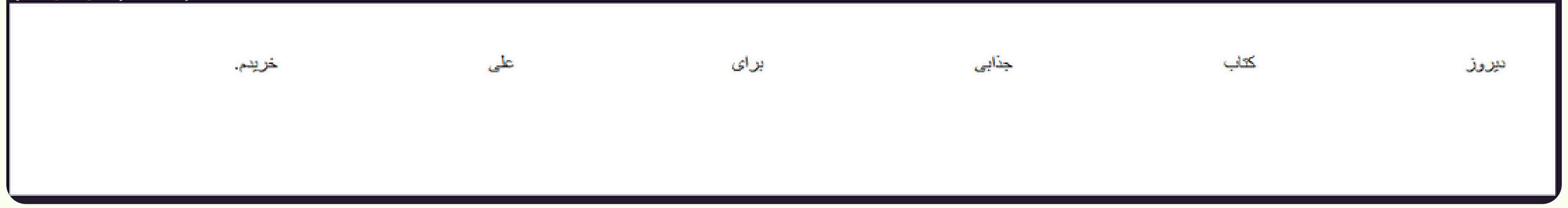
- Normalize text
- Tokenize+mwt
- POS+Morph tags
- Dep-parse model inference
- Visual / post-rules (ezāfe arcs, copula fill-in)

Capturing **head-dependent links** sharpens sentence understanding, boosting relation extraction, semantic search, and question answering.

```python
import stanza, spacy
# Stanza pipeline with GPU
nlp = stanza.Pipeline('fa', processors='tokenize,mwt,pos,lemma,depparse', use_gpu=True)
doc = nlp("دیروز کتاب جذابی برای علی خریدم.")

# spaCy visual (RTL)
nlp_sp = spacy.blank("fa_udpipe")
# nlp_sp.add_pipe("dependency_parser")
doc.sentences[0].print_dependencies()
spacy.displacy.render(nlp_sp(doc.text), style="dep", options={'direction':'rtl'})
```

```
('دیروز', 6, 'obl')
('کتاب', 6, 'obj')
('جذابی', 2, 'amod')
('برای', 5, 'case')
('علی', 6, 'obl')
('خریدم', 0, 'root')
('.', 6, 'punct')
```



https://paperswithcode.com/task/dependency-parsing

https://nlpdataset.ir/farsi/dependency_parsing.html

5. Name Entity Recognition (NER)

- Names often look like common words («گل»)
- No capitalization cues
- Honorifics & titles fused or abbreviated (دکتر اتابکی؛ آقای اکبری)
- Persian vs Arabic spelling variants («کاظمی / قاظمی»)
- Persian calendar, hijri & Roman dates

Entity categories

- PER (person)
- ORG (org)
- LOC (geo)
- FAC (facility)
- DATE
- TIME
- MISC

Key datasets

- PEYMA (248k tokens, news)–CC BY-NC
- ArmanPersoNER (70k tokens, blog/news)
- WikiFane 2024 (LM-ready Wikipedia dump)
- ParsGovNER 2024 (gov. docs, CC0)

```python
from transformers import pipeline

model_id = "HooshvareLab/bert-fa-base-uncased-ner-peyma"   # PEYMA-only
# model_id = "HooshvareLab/bert-fa-zwnj-base-ner"          # mixed ARMAN+PEYMA+WikiANN
ner = pipeline("ner", model=model_id, aggregation_strategy="simple")

text = "دکتر سارا گل‌محمدی در تهران سخنرانی کرد."
print(ner(text))

```

```
{'entity_group': 'B_PER', 'score': np.float32(0.999181), 'word': 'سارا', 'start': 5, 'end': 9}, {'entity_group': 'I_PER', 'sc
```

6. Sentiment Analysis Challenges

- No caps → entity ⇄ adjective ambiguity («عالی» can be a surname)
- Negation prefixes (نا-، بی) change scope late
- Sarcasm & intensifiers frequent in social text («چه خوب!»)
- Mixed scripts & emojis in tweets

Main datasets

- SentiPers (26k sents, aspect-level, digital products) GitHub
- DeepSentiPers (12k sents, 5-class & binary) Hooshvare Lab
- Digikala / SnappFood (e-commerce reviews) Hugging Face
- Pers-Twitter (3k tweets, 5 emotions) Kaggle

Models & lexicons

- HooshvareLab/bert-fa-base-uncased-sentiment-digikala (HF)
- m3hrdadfi/albert-fa-base-v2-sentiment-deepsentipers-multi
- ParsBERT v3 scores: 92 F1 (binary SentiPers), 88 F1 (SnappFood) GitHub
- Polarity lexicons: Dehkhoda ± lists, emoji valence tables

```python
from transformers import pipeline
sent = pipeline('sentiment-analysis',
                model='HooshvareLab/bert-fa-base-uncased-sentiment-digikala')

print(sent("این گوشی واقعاً بی‌نظیره!"))
```

```
[{'label': 'recommended', 'score': 0.9822326898574829}]
```

7. Persian Dataset Repositories

Category	Flagship resources (quick facts)
News & raw text	Hamshahri V2 – 323 k XML news stories, 1996-2007 (Kaggle, Wikipedia)
POS / Morphology	Bijankhan – 2.6 M manually-tagged words, 40 POS tags (Kaggle, ResearchGate)
Named-Entity corpora	PEYMA (302 k tokens, 7 labels) (Hooshvare Lab) ; ArmanPersoNER (250 k tokens, 6 labels) (GitHub)
Sentiment / Emotion	SentiPers (26 k sentences, aspect-level) (Hugging Face) ; SnappFood / Digikala review dumps (2024) (De Gruyter Brill)
Speech & audio	Common Voice fa v13 – 48 k validated clips (Hugging Face) ; FARSDAT – 300 speakers, 10 dialects (catalog.elra.info)
Massive pre-train	Matina – 72.9 B-token cleaned crawl (2025) (arXiv) ; Fars-Instruct / FarsGLM instruction-tuned mixtures (ACL Anthology)

8. Farsi NLP Libraries & Python Packages

Package	One-liner super-power / install hint
Hazm	Classic “Swiss-knife” for normalization, POS, dep-parse → pip install hazm (GitHub, PyPI)
Parsivar	Fast rule-based pre-processing + CRF POS/NER → pip install parsivar (PyPI, GitHub)
Stanza-fa	GPU-ready UD pipeline (token → dep parse) → stanza.Pipeline('fa') (Hugging Face, Stanford NLP)
DadmaTools	spaCy-based neural pipeline incl. ezāfe detector → pip install dadmatools (GitHub, ACL Anthology)
Hezar	“All-in-one” HuggingFace-friendly Persian AI toolkit → pip install hezar (GitHub, hezarai.github.io)
spaCy-fa (alpha)	Lightweight blank pipeline for custom model drop-ins → spacy.blank("fa") (GitHub)

https://github.com/roshan-research/hazm

You can also explore **PersianLLaMA** and fine-tuned **MT5-base** for generation.

9. Future Directions (2025-26)

Trend 2025-26	What to watch
Large-scale Persian LLMs	Matina 72.9 B-token corpus powering GPT-4-class open models; bilingual PersianMind built on LLaMA-2 keeps English skills while excelling in Fars
Instruction-tuning & alignment	FarsInstruct dataset (3 M dialogs) pushes chat quality; RLHF and policy-tuning now feasible in Persian. (arXiv)
Multimodal Persian models	“Persian in a Court” benchmark tests OCR-QA & VQA; first vision-language checkpoints due late 2025. (ACL Anthology)
Cross-dialect & code-switch	Telegram & social-media corpora expand coverage of Dari, Tajik, Pinglish; cross-lingual pre-train bridges EN-FA gaps. (Hugging Face)
Open speech boom	ManaTTS (100 h), GPTInformal speech pairs, and Filimo ASR movies fuel next-gen TTS/ASR. (GitHub, GitHub, Hugging Face)
Unified tooling & benchmarks	Hezar library unifies pipelines; refreshed ParsGLUE-v2 due mid-2025 with NLI (FarsTail), QE (EPOQUE). (GitHub, Papers with Code, ACL Anthology